Discourse-JEPA: Latent World Models over Reasoning Steps

Historical experiment chronology (archival)

Paper-facing replacement: `discourse_jepa_neurips.pdf`

TextJEPA project

July 13, 2026 (repo: `TextJEPA/`, runs: `runs/`)

Idea in one slide

- Treat a **reasoning trace** as trajectories of a world model: the *state* s_t is a compressed discourse state ("what has been established"), the *action* a_t is a tiny latent code of an *intent phrase* for the next step.
- Actions state intent only (*"derive purple apples from orange jars times green pens ."*) — never the outcome ("$= 5$"). The predictor $F(s_t, a_t)$ must therefore **compute consequences in latent space**.
- Generation = **search over actions**: roll candidate actions forward with F, score with an energy, execute the best in the environment, re-encode, replan (closed-loop MPC). The world model predicts **latents, not outcome tokens**; an optional training-only LDAD head reconstructs the observed action phrase and is discarded at planning time.
- Synthetic reasoning world $\Rightarrow$ exact ground truth for **linear probes** and **objective planning success**.

Research questions and current answers

1. **A latent world model over reasoning steps works** — and we can *audit* it: predicted futures of never-taken actions match symbolic execution (Results 1, 2, 9).
2. **What silently breaks**: jointly learned encoders can discard computed content to simplify prediction (Result 3); latent distance is not a progress metric (Result 5).
3. **Reconstruction-free fixes**: frozen random-init embedding targets restore content (Result 4); trajectory-geometry losses fix the metric (Results 7, 8); counterfactual *ranking* fixes the energy's discrimination (Result 10).
4. **The energy can come from the geometry itself**: a value head distilled from latent goal distance, without scalar distance-to-go labels, matches the scalar-supervised alternative within seed variance (Results 13, 16).
5. **Baselines and limits**: likelihood selection (LMs at any tested scale/granularity) clusters far below (Result 15); failures are chain-depth-limited — the scale-up frontier (Result 16).

Notation and experimental terminology

discourse state s_t	one vector summarizing prompt + steps $1..t$ (output of the causal state model)
predictor $F(s, a)$	MLP producing the next state from state + action code (the world model)
action a	16-d encoding of an *intent phrase* ("derive X from Y times Z") — states what to do, never the outcome
value head $V(s, s_0)$	scalar energy: estimated necessary steps remaining; the planner minimizes steps $+ V$
end-to-end value shaping	value-head gradients enter the encoder so the planning energy can shape the representation
observed-action displacement decoding	reconstruct the complete observed intent-token sequence from $\Delta s = s_{t+1} - s_t$; our historical hybrid operation/embedding target is reported only as a distinct control
fixed outcome target	predict a frozen random-feature embedding of the next sentence; prevents loss-minimizing content deletion
counterfactual preference loss	margin loss ordering $V(F(s, a_i))$ over K counterfactual actions per state by their symbolic outcomes
cross-horizon cost calibration	order multi-step costs (depth $+ V$) — calibrates search depth
goal-monotonicity constraint	necessary steps must decrease the latent distance to the trace-terminal state by a margin
geometry-distilled value	value head regressed on that latent goal distance (geometry target, no numeric labels)

Scientific labels and artifact identifiers

Artifact names are retained only for reproducibility; the report uses the scientific labels in the left column.

scientific label	repository artifact
fixed-outcome prediction + end-to-end value shaping	`disc_combo`
goal-monotone value regression	`disc_mono_hi`
counterfactual preference learning ($K = 2$)	`disc_rank_k2`
geometry-distilled counterfactual model	`disc_mono_distill_rank`
five-objective ranked JEPA	`disc_core5`
cross-horizon calibrated model	suffix `_cal`
independent seed replicates	suffixes `_s2`, `_s3`

The five-objective model uses one-step latent prediction, counterfactual preference learning, VICReg, fixed outcome targets, and displacement decoding. The "full" model additionally uses rollout, hierarchy, and goal-geometry terms.

Headline comparison: fixed one-step planning protocol

Claim: Under identical one-step MPC, the five-objective ranked JEPA improves strict-budget success by 14.5 points over the largest token LM tested.

	@optimal	@slack-2
random feasible policy	0.055	0.405
token LM, parameter-matched (best lr)	0.670	0.910
token LM, 3$\times$ larger	0.735	0.940
five-objective ranked JEPA	**0.880 $\pm$ 0.004**	**0.975 $\pm$ 0.004**
geometry-distilled ranked JEPA	0.900 $\pm$ 0.027	0.980 $\pm$ 0.008
symbolic oracle	1.00	1.00

All learned rows use greedy lookahead 1 and the same 200 validation episodes. The five-objective model's value head is trained by pairwise margins, without scalar remaining-step labels. Mean $\pm$ standard deviation uses three seeds.

Search-depth interaction is a separate experiment

Claim: Cross-horizon calibration helps the full model, but not the five-objective model; it is not a generic "depth option."

training objective	lookahead 1	lookahead 2	slack-2, lookahead 2
five-objective + cross-horizon calibration	0.900	0.885	0.985
full geometry-distilled + cross-horizon calibration	0.925	**0.945**	**1.000**

The five-objective model combines latent prediction, fixed outcome prediction, VICReg, displacement decoding, and pairwise preferences. The full geometry-distilled model additionally shapes goal-distance monotonicity and distills that distance into V. Single-seed interaction test. "Cross-horizon" means that training compares $1 + V(F(s,a))$ with $2 + V(F(F(s,a),a'))$; "lookahead" is the matching planning-time rollout depth. This is not part of the three-seed headline table.

Value-head supervision targets compared (symbolic vs geometric)

Claim: With matched pairwise preference supervision, geometry-distilled values match scalar-supervised values within seed variance, provided the geometry is goal-monotone.

V trained on	metric shaping	labels used	@opt	@slack2
numeric remaining-steps	—	scalar counts	0.64±.05	0.89
numeric + ranking	—	counts + order	0.91±.03	0.99
numeric + ranking + hinge	hinge	counts + order + binary	0.89±.02	0.99
latent goal distance	**none** (control)	none	0.19	0.64
latent goal distance	curvature (label-free)	none	0.30	0.72
latent goal distance	hinge	binary relevance	0.57±.05	0.89
lat. dist. + ranking	hinge	binary + order	**0.90±.03**	**0.98**

Reading: the distilled (geometric) value head is a drop-in replacement for numeric supervision once the metric it distills is goal-monotone; scalar count labels contribute nothing beyond binary relevance + ordering (0.90 vs 0.89/0.91, within seed noise).

World: iGSM-style synthetic reasoning (exact generator)

Problem sampler (rejection-sampled until $3 \leq |\mathcal{A}_q| \leq 9$; `rng = Random(f"{seed}:{index}")`):

- $n \sim \mathcal{U}\{6, \ldots, 12\}$ variables; names $=$ distinct (adjective, noun) pairs from 20×20 vocabulary.
- Variable i: with p=0.35 (always for $i < 2$) a **leaf** $x_i = c_i, \; c_i \sim \mathcal{U}\{0..22\}$; else an **internal** $x_i = x_a \circ x_b$ mod 23, $\circ \in \{+, -, \times\}$, parents a, b sampled from $\{0..i-1\}$.
- Query q $=$ uniform over internal variables; $\mathcal{A}_q$ $=$ ancestors of q (incl. q) $=$ the necessary computation; the rest are **distractors**.

Training traces: solve by repeatedly resolving a feasible variable (parents resolved); with p=0.15 (max 2) pick a feasible *distractor* instead of a necessary one $\Rightarrow$ off-path states for the value head. **Important protocol correction.** Published artifacts in this deck reused the same deterministic index set each epoch. New runs (July 13 onward) use disjoint epoch-offset indices, preserving determinism while generating fresh problems; validation remains fixed.

A verbatim sample (val problem #3)

Prompt (definitions shuffled per-sample, question last; one sentence = one chunk):

the number of purple apples equals the number of orange jars times the number of green pens .
the number of orange jars is 15 . *the number of green pens is 8 .*
(+ 7 distractor definitions) *how many purple apples are there ?*

Solution trace — step sentence (observed outcome) vs. action intent phrase (no outcome):

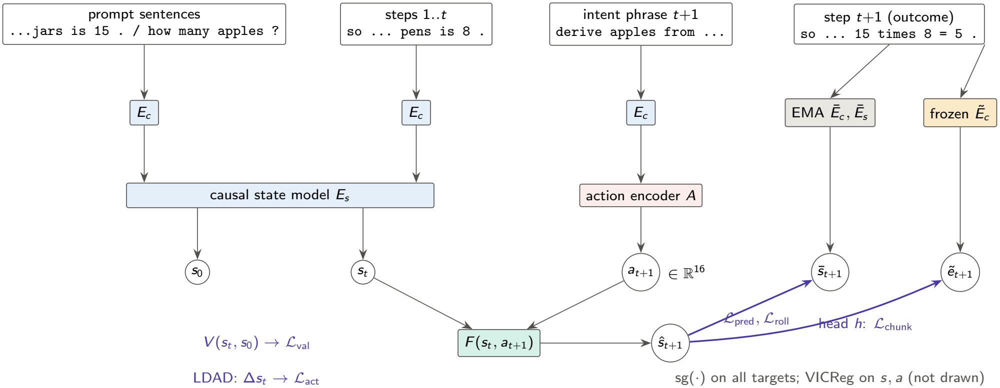
step sentence (encoder sees)	intent phrase (action encoder sees)
so the number of green pens is 8 .	look up the number of green pens .
so the number of orange jars is 15 .	look up the number of orange jars .
so the number of purple apples is 15 times 8 = 5 .	derive purple apples from orange jars times green pens .

Note $15 \times 8 = 120 \equiv 5 \pmod{23}$: outcomes are nontrivial to predict.

Architecture (all sizes exact; 8.8–9.1M params)

- **Chunk encoder** E_c: word-level vocab (441 tokens), embed 256, 2-layer pre-norm transformer (4 heads, FF 1024), masked mean-pool $\rightarrow$ one 256-d vector per sentence. Shared for prompt, steps, intents.
- **State model** E_s: 4-layer *causal* transformer (8 heads, FF 1024) over [prompt chunks | step chunks] + learned positions + 2 segment embeddings. $s_0 =$ hidden at the question; $s_t =$ hidden at step t.
- **Action encoder** A: MLP 256 $\rightarrow$ 256 $\rightarrow$ 16 on the intent-phrase embedding (optional FSQ, unused in main runs).
- **Predictor** F: residual MLP, $\hat{s}_{t+1} = s_t + \mathrm{MLP}([\mathrm{LN}(s_t); a_t])$, 2 hidden layers of 1024 (GELU).
- **Hierarchy**: CLS-transformer over $K{=}3$ action codes $\rightarrow$ 8-d macro action m_t; $F_{\mathrm{hi}}(s_t, m_t)$ predicts $\bar{s}_{t+3}$ (same latent space, HWM-style).
- **Observed-action displacement decoder**: position-query transformer on $\Delta s = s_{t+1} - s_t$ reconstructs the complete intent phrase. The historical hybrid head (operation class + EMA intent embedding) is not a faithful Delta-JEPA target.
- **Value head** V: MLP on $[\mathrm{LN}([s_t; s_0])] \rightarrow$ scalar $=$ predicted remaining necessary steps.
- **Teachers**: EMA copies of E_c, E_s (momentum 0.99 $\rightarrow$ 0.999 linear); **frozen anchor** $\tilde{E}_c =$ copy of E_c at random init, never updated.

Training pipeline



Objectives (exact losses and weights)

Let $d(x, y) = \mathrm{smoothL1}(\mathrm{LN}(x), \mathrm{LN}(y))$ (per-dim mean; LN = parameter-free layer norm), masks over valid steps.

$$
\begin{aligned}
\mathcal{L}_{\text{pred}} &= d\big(F(s_t, a_t),\, \mathrm{sg}\,\bar{s}_{t+1}\big) && \text{weight } 1.0 \\
\mathcal{L}_{\text{roll}} &= d\big(F^{(t)}(s_0, a_{1:t}),\, \mathrm{sg}\,\bar{s}_t\big) \ \text{(open loop, all } t\text{)} && 1.0 \\
\mathcal{L}_{\text{hi}} &= d\big(F_{\mathrm{hi}}(s_t, m_t),\, \mathrm{sg}\,\bar{s}_{t+3}\big) && 0.5 \\
\mathcal{L}_{\text{vic}} &= \textstyle\sum_j \max(0, 1 - \mathrm{std}(s^{(j)})) + 0.04\,\|\mathrm{offdiag}(\mathrm{Cov}(s))\|_F^2/D + 0.1\,\mathrm{var}(a) && 1.0 \\
\mathcal{L}_{\text{obs-act}} &= \mathrm{CE}\big(D(\Delta s_t),\, \mathrm{tokens}(u_t)\big) \quad \text{(faithful observed-action LDAD)} && 0 \ / \ \text{selected} \\
\mathcal{L}_{\text{val}} &= \mathrm{smoothL1}\big(V(s_t, s_0),\, \#\text{unresolved necessary}\big) && 1.0 \\
\mathcal{L}_{\text{chunk}} &= d\big(h(\hat{s}_{t+1}),\, \mathrm{sg}\,\tilde{E}_c(y_{t+1})\big) + 0.5\, d\big(h(\text{rollout}_t),\, \mathrm{sg}\,\tilde{E}_c(y_t)\big) && 0 \ / \ 2.0 \\
\mathcal{L}_{\text{curv}} &= 1 - \cos(v_t,\ v_{t+1}), \quad v_t = s_{t+1} - s_t \ \text{(straightening, arXiv:2603.12231)} && 0 \ / \ .02\text{–}.1 \\
\mathcal{L}_{\text{mono}} &= \nVdash_{\text{nec}} [\Delta d_t^{\text{goal}} + m]_+ + \tfrac{1}{2}(1 - \nVdash_{\text{nec}})\, [-\Delta d_t^{\text{goal}}]_+ && 0 \ / \ 1\text{–}2
\end{aligned}
$$

$\mathcal{L}_{\text{chunk}}$: weight 0 in `base`; 2.0 in `chunkpred`/`combo` (**frozen** anchor $\tilde{E}_c$). Value head input detached except `valgrad`/`combo`. d_t^{goal} = LN-L1 distance of s_t to the trace-terminal EMA state; margin $m \in \{.02, .05\}$.

Training protocol

- AdamW, lr 3×10^{-4}, weight decay 0.05 (none for dim < 2 params), $\beta = (0.9, 0.95)$; cosine schedule to floor 0.05, 1000-step warmup; grad clip 5.0.
- Batch 128 problems; 20 epochs $\times$ 781 steps at the 100k setting. Historical results reused a deterministic 100k corpus. Future runs use a deterministic disjoint corpus per epoch. EMA momentum $0.99 \to 0.999$ linear over training.
- Hardware: one V100-32GB, fp32, $\sim$40 min/run.
- In-training eval each epoch: all losses + collapse diagnostics (per-dim std, RankMe effective rank) + closed-form ridge probes.
- Code: `scripts/train.py +experiment=...`; every run's exact config is stored in its checkpoint.

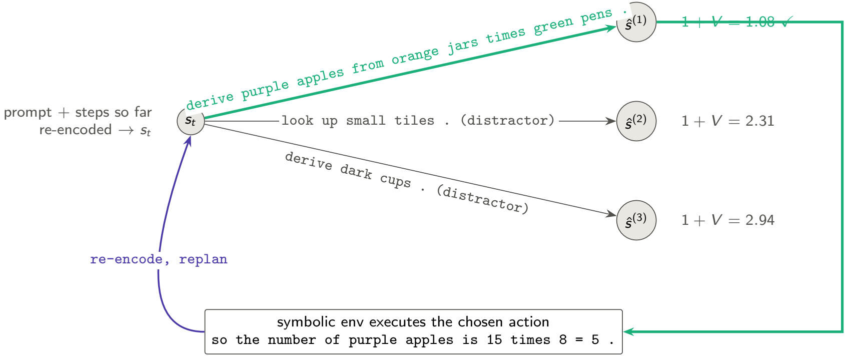
Planning: closed-loop MPC by exhaustive latent search

Interface knowledge (legitimate): feasible actions = unresolved variables whose stated dependencies are resolved — readable from the prompt. **Consequence knowledge**: latent only.

1. Encode prompt + steps so far $\rightarrow s$ (and s_0 once).
2. Enumerate feasible action *sequences* to depth L (lookahead; cap 64), encode their intent phrases $\rightarrow a_i$.
3. Roll out $\hat{s} \leftarrow F(\hat{s}, a_i)$ along each sequence.
4. Score: $\text{cost} = \text{steps taken} + V(\hat{s}_{\text{end}}, s_0)$ (estimated total steps).
5. Execute the first action of the argmin in the symbolic environment $\rightarrow$ real outcome sentence appended; go to 1.

Success = query resolved within the **strict optimal budget** $|\mathcal{A}_q|$ (slack 0: any distractor pick is fatal). 200 val episodes. Discrete enumeration replaces CEM; a sampled action prior would need CEM/MPPI.

Planning: search graph (one replanning round, lookahead 1)



V values are predicted *remaining necessary steps* of the predicted next latent; lookahead 2 scores 2-step latent rollouts.

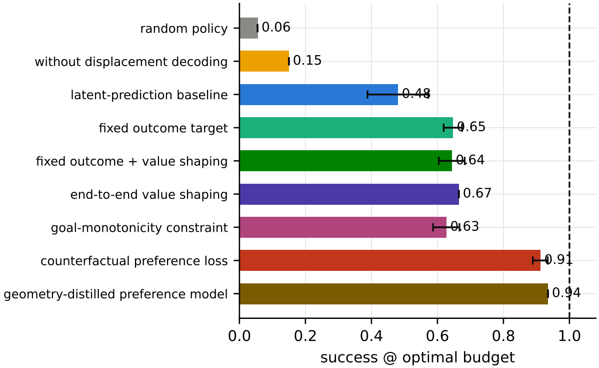
Experiment grid (each condition is one 20-epoch run)

condition	fixed outcome target	value grad $\rightarrow$ encoder	additional change
latent-prediction reference	—	no	—
without displacement decoding	—	no	$\mathcal{L}_{\text{act}}$ weight 0
fixed outcome prediction	frozen, w=2.0	no	—
end-to-end value shaping	—	**yes**	—
fixed outcome + value shaping	frozen, w=2.0	**yes**	—
value-only control	—	**yes**	*all* JEPA weights 0
frozen-encoder control	—	no	encoders frozen at init
curvature-weight sweep	frozen, w=2.0	**yes**	$+\mathcal{L}_{\text{curv}}$, $\lambda \in \{.02, .05, .1\}$
goal-monotonicity sweep	frozen, w=2.0	**yes**	$+\mathcal{L}_{\text{mono}}$, w1 m.02 / w2 m.05
combined geometric constraints	frozen, w=2.0	**yes**	both regularizers
monotonicity without value head	frozen, w=2.0	no	$+\mathcal{L}_{\text{mono}}$, *no value head*
counterfactual preferences ($K = 2, 4$)	frozen, w=2.0	**yes**	+ ranking over K counterfactual actions
preference + goal monotonicity	frozen, w=2.0	**yes**	ranking K=2 + $\mathcal{L}_{\text{mono}}$ w2
grounding and data controls	frozen, w=2.0	**yes**	grounding controls (Results 9, 11)

Baselines at plan time: **random feasible policy**, **symbolic oracle** (=100%), and **oracle goal-distance energy** (LeWM-style, uses the encoded solved state — diagnostic only).

Result 1 — historical objective sweep (not an ablation shear)

Claim: Fixed outcome targets improve the latent baseline; counterfactual preference learning gives the largest further gain. Adjacent bars are separate recipes and must not be read as a cumulative one-factor-at-a-time sequence.



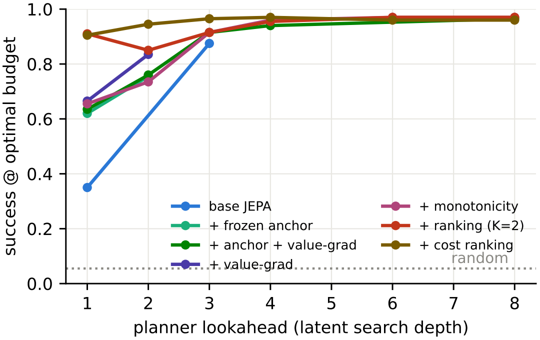
Success within the *strict optimal budget*, greedy lookahead 1, 200 episodes, mean necessary steps 4.07.

With slack 2 (mean of 3 seeds): latent baseline 0.83, fixed outcome target 0.89, goal monotonicity 0.91, counterfactual preference models **0.98–0.99** (random 0.41).

Distractor pick rate: random 0.43 → latent baseline 0.28 → end-to-end value shaping 0.14. The subtractive/constructive component tests appear in Result 16.

Result 2 — search depth is a planning-time compute knob

Claim: Success grows with search depth up to a shared ≈0.97 ceiling: rollouts of F and the energy stay trustworthy over many imagined steps, and depth even rescues weak energies.



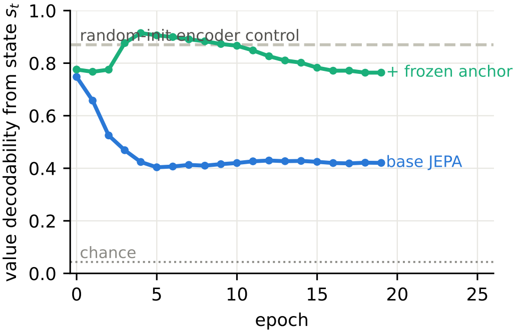
Lookahead d = feasible d-step sequences (capped), scored at the imagined end state; no retraining. Depth curves are single-seed; the depth-1 anchors carry ±0.03–0.05 seed bands (leaderboard slide).

All recipes converge to ≈0.96–0.97 by depth 4 — the residual is the deep-chain failure mass (Result on failures), not search.

Base JEPA: 0.35 → 0.87 (depth 3); `mono_hi` 0.655 → 0.97 (depth 6). The `rank_k2` dip at depth 2 is the calibration anomaly; cost ranking (brown) removes it and dominates at every depth.

Result 3 — encoder–predictor co-adaptation, measured

Claim: Pure state-prediction JEPA *removes* outcome information that was linearly present at initialization; a frozen outcome anchor prevents it.



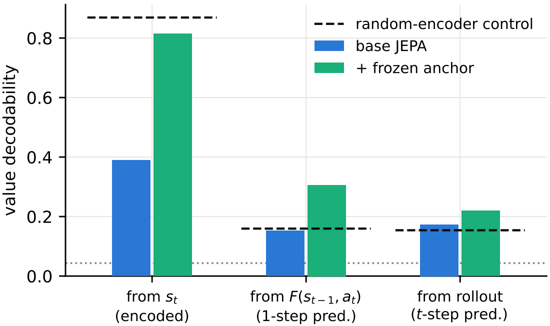
Probe: linear decodability of the just-computed value (23 classes) from s_t, val split.

Both loss sides are learned; EMA targets *trail the online encoder*, so deleting a hard-to-predict feature lowers the loss — stop-grad does not pin content.

Consequence in base: answer decodable from the terminal state at only 0.26 (random-init control: 0.65).

Result 4 — latent arithmetic under a fixed outcome target

Claim: With $\mathcal{L}_{\text{chunk}}$, the predictor computes outcomes it never observes: value decodability from *predicted* latents doubles the random-encoder control.



Dashed = random-init encoder control (surface memory, no computation); dotted = chance $1/23$.

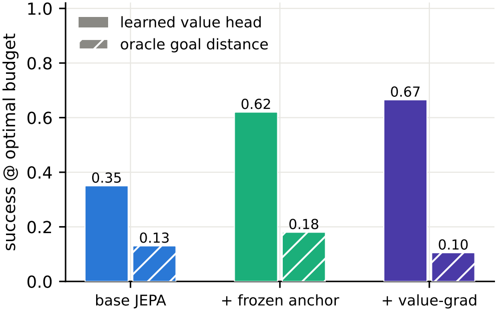
From $F(s_{t-1}, a_t)$: baseline 0.15 (= control) $\rightarrow$ fixed outcome target **0.31**.

Answer from terminal state: 0.26 $\rightarrow$ **0.98**.

Full-trace open-loop rollout: 0.22 (error compounds; matches Result 2's need for re-encoding).

Result 5 — the goal energy must be learned (LeWM test)

Claim: Raw latent distance to the oracle goal state plans barely above random; the trained value head is the energy that works.



Oracle goal = encoded terminal state of the *true* solution (planner gets it for free — upper-bound diagnostic).

Scoring $\|\mathrm{LN}(\hat{s}) - \mathrm{LN}(s_{\mathrm{goal}})\|_1$: latent baseline 0.13, fixed outcome target 0.18, end-to-end value shaping 0.105 (distractor rates $\approx$ random).

$\Rightarrow$ *unregularized* JEPA geometry is not a distance-to-goal metric. Result 7 shows temporal straightening fixes exactly this.

Result 6 — displacement decoding ablation; stability

Claim: Our hybrid displacement decoder is useful for planning, but this is not an exact replication of Delta-JEPA's raw-action reconstruction objective.

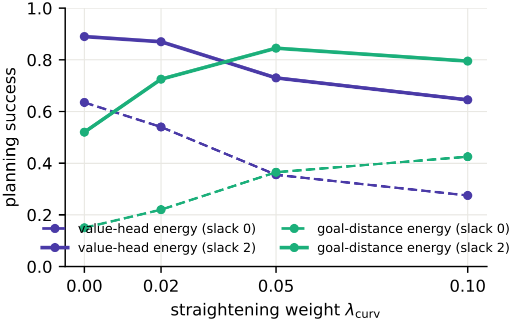
	success@opt	success@slack2	distractor rate	op from Δs
`base`	**0.35**	**0.75**	0.28	1.00
`no_delta`	0.15	0.51	0.40	0.98*

*Operation type stays decodable (it correlates with surface features) but transitions lose the *action-discriminative* structure the planner needs.

Stability (all runs): per-dim state std ≈ 1.03, RankMe effective rank 220–243 / 256, no collapse; VICReg + EMA sufficed, discrete codes (FSQ) not needed. Delta-JEPA decodes the observed continuous action from $s_{t+1} - s_t$ with MSE and trains end-to-end without EMA or a distribution regularizer. Here the targets are an operation label and an EMA intent embedding; operation CE is therefore explicit symbolic action-type supervision.

Result 7 — temporal straightening makes raw geometry plannable

Claim: The curvature loss (arXiv:2603.12231) turns latent distance into a progress metric: goal-distance planning goes 0.15 → 0.43 @optimal (0.52 → 0.85 @slack 2) — but it *trades off* against the value head.



Dose-response over $\lambda_{\text{curv}} \in \{0, .02, .05, .1\}$ (all on the fixed-outcome, end-to-end value model):

	0	.02	.05	.1
velocity cos	$-.25$	$-.01$	.42	.67
corr(d^{goal}, remaining)	.73	.75	.80	.86
value probe from s_t	.89	.89	.74	.43

Geometry improves exactly as content decodability degrades; $\lambda = .05$ is the raw-geometry sweet spot.

Result 8 — two energies, one metric: the content–geometry trade-off

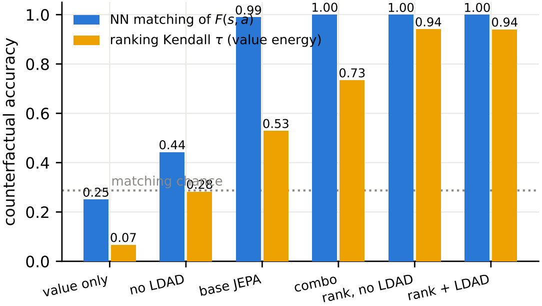
Claim: Straightening buys geometry planning, the goal-monotonicity constraint keeps content and sets the value-planning record; geometry needs JEPA losses, not value labels.

run	goal-distance energy		value-head energy	
	@opt	@slack2	@opt	@slack2
`combo` ($\lambda = 0$)	0.15	0.52	0.635	0.89
`straight_mid` (λ=.05)	0.365	**0.845**	0.355	0.73
`mono_hi` (hinge w2 m.05)	0.255	0.68	**0.655**	**0.935**
`mono_novalue` (*no V*)	0.20	0.65	—	—
`value_only` (*no* JEPA)	0.06	—	0.075	0.43
random	0.06	0.095	0.06	0.095

Three readings. (i) One latent metric cannot currently maximize both energies — a learned *projection head* for the geometric energy is the obvious fix (tested in Result 10: it softens but does not remove the trade-off). (ii) A model with **no value head** still plans at 0.65 @slack 2 from pure geometry. (iii) `value_only`'s geometry is *exactly random*: planning-relevant structure comes from the JEPA objectives, not the value labels.

Result 9 — the predictor is counterfactually grounded (direct audit)

Claim: From each state we enumerate *all* feasible actions, predict their next-state latents, and check against symbolic execution the model never saw: $F(s, a)$ is correct per action, and displacement decoding / fixed outcome targets are what ground it.



Audit metrics (100 episodes, val): nearest-neighbor *matching* of $F(s, a_i)$ to the executed-and-encoded true next states; Kendall τ of the value energy vs. true post-action remaining steps.

Training data shows **one action per state** — grounding comes from cross-problem generalization alone.

Shuffled-action control: matching 0.30 $\approx$ chance. RSA of pairwise next-state geometry: 0.93 (base+LDAD) vs 0.48 (no LDAD).

Result 10 — counterfactual ranking: the strongest lever

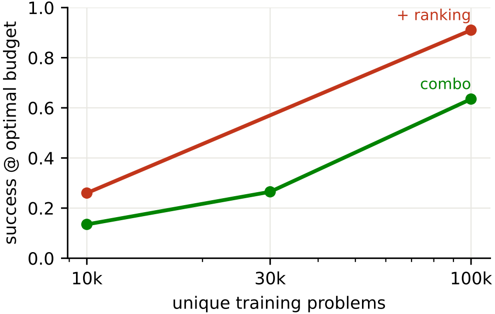
Claim: Hinge-ranking K alternative actions per state by symbolic outcome nearly closes the gap to the oracle; K=2 suffices; it can replace LDAD but not data diversity; it trades search-depth calibration for 1-step precision.

	@opt	@slack2	@opt look-2	value τ
`combo`	0.635	0.89	0.76	0.73
`rank_k2`	0.910	0.985	0.85 ↓	0.94
`rank_k4`	0.925	0.98	0.85 ↓	0.94
`rank_k4` *no LDAD*	0.875	0.965	—	0.94
`combo` *no LDAD*	0.48	0.805	—	0.69
preference + goal monotonicity	**0.905**	**0.99**	—	**0.96**

↓ lookahead-2 *hurts* ranking models (0.91→0.85) while helping regression values (mono_hi 0.655→0.735): the margin loss perfects 1-step ordering but distorts V's absolute scale that multi-step cost sums need. Preference learning + goal monotonicity also has the best raw geometry (τ_{goal} 0.82, oracle-goal planning 0.655, no curvature loss). Seed replicates: rank_k2 0.885–0.91 @opt, 0.98–0.985 @slack2.

Result 11 — data-scale trend (coverage and compute confounded)

Claim: Performance increases with the procedural corpus size, but the original sweep also changes optimizer updates; it is evidence for a scale trend, not a clean causal estimate of data diversity.



Fixed 20 epochs: 10k, 30k, and 100k imply approximately 1.6k, 4.7k, and 15.6k optimizer updates, respectively.

problems	10k	30k	100k
combo @opt	0.135	0.265	0.635
+ ranking	0.26	—	0.91
matching	0.82	1.00	1.00
value τ	0.22	0.44	0.73

Matching survives small data; *ranking quality* does not — the predictor stays roughly right, the energy loses discrimination. Within-trace variation matters too: no distractors $\Rightarrow$ 0.635$\rightarrow$0.415. A compute-matched fresh-stream rerun is now required.

Result 12 — cross-horizon calibration and macro transitions

Claim: Plain ranking breaks lookahead (0.91→0.85); two independent fixes exist, and they don't stack because they repair the same defect.

	look-1	look-2	look-3 flat	look-3 F_{hi}
`rank_k2`	0.910	0.850 ↓	0.915	**0.970**
`rank_cal` (+ cross-horizon calibration)	0.905	**0.945**	—	0.915
preference + monotonicity + calibration	0.910	0.945	—	0.915

Cross-horizon calibration: rank full MPC costs across depths — executed 2-step continuation $(2 + V(F^2))$ vs 1-step alternatives $(1 + V(F(a')))$; zero new data. **Macro jumps**: score 3-step sequences with one F_{hi} call — worse *fidelity* than F^3 (matching 0.62–0.70 vs 0.77–0.91) but stays on V's training distribution. 1000-episode: rank_cal look-2 = 0.946 vs hier 0.931 — calibration wins once you have it; macro jumps are the no-retraining workaround. Train-side hierarchy loss: neutral (`combo_nohier` 0.670/0.910).

Result 13 — the non-symbolic value head (headline)

Claim: Shape the metric with goal monotonicity, distill it into V, add preference learning: best planner with no numeric value labels anywhere.

run	energy supervision	@opt	@slack2
scalar value + preference + calibration	*numeric* + order	0.91 ± .01	0.995
`mono_distill_rank`	**binary only (geometric V)**	**0.90 ± .03**	**0.98 ± .01**
`mono_hi`	*numeric* regression	0.63 ± .05	0.91 ± .02
`mono_distill`	binary (no ranking)	0.57 ± .05	0.89
`mono_lf_distill`	*none* (all-steps hinge)	0.355	0.775
`geometric`	*none* (curvature)	0.300	0.715
`distill_v`	*none, unshaped* (control)	0.190	0.640

Rows 1 vs 2 are the direct **numeric-vs-geometric** comparison at matched search settings: the geometric value head (V regressed on the latent goal distance d^{goal}, ordered by binary-outcome ranking) matches the scalar-label comparison within seed noise (0.90±.03 vs 0.91±.01). Shaping is required: unshaped distillation fails (0.19). Mean±std over 3 seeds where shown; single-seed otherwise.

Result 14 — stability of the hybrid displacement objective

Claim: Our operation/intent displacement decoder alone does not prevent collapse; VICReg remains important in the observed-intent architecture. This does not test Delta-JEPA's exact raw-action objective.

run	EMA	sg	VICReg	result
`ldad_pure`	—	—	—	**collapse** (std .027, rank 80)
`ldad_pure_sg`	—	✓	—	collapse (0.11 @opt)
`combo_novic`	✓	✓	—	0.27 @opt (vs 0.635)
`combo`	✓	✓	✓	0.635 @opt

This is a negative result for our hybrid decoder, not a negative replication of arXiv:2606.31232: targets, encoders, and action observability differ. Also: non-residual predictor $\hat{s} = \mathrm{MLP}([\mathrm{LN}(s); a])$ performs at least as well as residual (0.710/0.920) — the residual skip is convenience, not mechanism; LDAD constrains *encoder* displacements, not predictor outputs, so it is not trivialized by either choice.

Result 15 — historical outcome-candidate LM diagnostics

Claim: Outcome-likelihood/reconstruction selectors cluster at 0.47–0.74; these are strong controls, but their candidate sentences expose computed consequences and are not information-matched policy baselines.

model (selection energy)	params	@opt	@slack2
sentence-LM (decoder CE)	7.2M	0.470	0.810
sentence-LM+latent tgt (decoder CE)	7.2M	0.510	0.800
sentence-LM+latent tgt (**latent dist.**)	7.2M	0.605	—
token LM (likelihood, best lr)	8.2M	0.670	0.910
naive JEPA (combo, value head)	9.1M	0.635	0.890
token LM	25.5M	0.735	0.940
cross-horizon calibrated JEPA	9.1M	**0.935**	**0.99–1.00**

All models enumerate the same feasible actions and use the same budget, but the historical LMs score rendered outcomes whereas JEPA scores outcome-free intent phrases. Information-matched intent-policy LMs are running. LR was swept for both families. Note the within-model dissociation: the sentence-LM's *latent* energy outranks its own decoder (0.605 vs 0.510) — selection is a latent-geometry problem even for reconstruction-trained models.

Result 16 — objective ablations

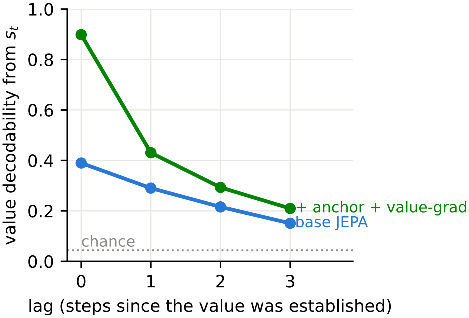
Claim: A constructive sequence identifies anti-collapse and predictive ingredients; independent removals from the full model localize auxiliary roles. These are two analyses, not one cumulative "recipe" curve.

constructive sequence	@opt	eff. rank	**removed from full model**	Δ@opt
ranking only	0.20	**24 (collapse)**	hinge	-0.02
+ latent prediction	0.78	101	distilled V	-0.04
+ VICReg	0.745	234 (healthy)	rollout supervision	~ 0
+ fixed outcome + displacement decoding	**0.88±.004**	234	hierarchy loss	~ 0
full geometry-distilled model	0.90±.03	233	EMA teacher	-0.03
			scalar value labels	~ 0
			residual skip	$+0.02$

Auxiliary roles remain identifiable: goal monotonicity + geometry-distilled V defines the non-scalar-value variant (0.36–0.57); EMA = +0.03 over stopgrad+VICReg (which suffice for stability — `noema` 0.605, no collapse); cross-horizon ranking calibrates depth, but only with the full objective set (five-objective: 0.885 look-2 $<$ 0.900 look-1; full: 0.945). **Failures remain chain-depth-limited**: 4.4% (3–4 steps) $\rightarrow$ 59% (7–9). The edit track shows the same ablation: the five-objective model reaches 0.455/0.485, matching the full model without defect labels.

Probes I — what the discourse state actually is

Claim: A recency-weighted working memory over a relational scaffold: structure is objective-independent, content is objective-dependent.

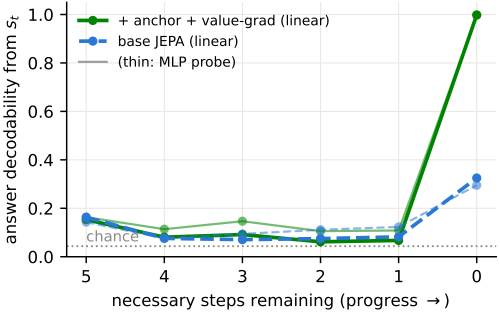


probe (combo / all recipes)	acc	maj.
resolved-set member $[s_t; \mathrm{oh}(v)]$	0.82	0.67
ancestor-of-query $[s_0; \mathrm{oh}(v)]$	0.76	0.56
var just resolved (from Δs)	0.33	0.19
query identity (from s_0)	0.19	0.16

These rows are **identical across every training recipe** (base, no-LDAD, geometry, ranking) while value decodability swings 0.39–0.90: the encoder learns the scaffold from text alone; the objectives decide what content survives. Relevance is encoded *relationally* (0.76), not as a query pointer (0.19).

Probes II — emergence, memory, circular code

Claim: No anticipatory answer computation: the answer appears exactly when its variable resolves — and only collusion-free models can hold it.



Emergence is a step function: 0.06–0.18 mid-trace (thin MLP lines: small nonlinear surplus), 1.00 the moment the query resolves; base manages only 0.33 even then (collusion).

Memory decays with lag (Probes I): 0.90 $\rightarrow$ 0.43 $\rightarrow$ 0.29 $\rightarrow$ 0.21 (chance 0.04) — a working buffer, not a symbol table.

Circular coding: ridge R^2 for $\cos(2\pi v/23)$ from s_t: 0.52 (combo) vs 0.20 (base) — the mod-p value is partly geometric, and the circle sharpens under the anchor fix.

Probe suite (fixed outcome target vs. baseline and random encoder)

probe (linear, val split)	`base`	`chunkpred`	random-enc
value of current step from s_t	0.39	**0.81**	0.87
value from $F(s_{t-1}, a_t)$ (latent arithmetic)	0.15	**0.31**	0.15
value from open-loop rollout	0.17	**0.22**	0.15
op type from Δs	1.00	1.00	0.97
step goal-relevance from Δs	0.89	**0.93**	0.89
remaining steps from s_t	0.44	**0.61**	0.41
resolved count from s_t	0.97	0.92	0.95
final answer from terminal state	0.26	**0.98**	0.62
answer from prompt alone s_0 (should be hard)	0.035	0.05	0.04
value-head MAE (remaining steps)	0.76	0.57	—

End-to-end value shaping gives MAE **0.38** with the same co-adaptation-resistant probes (state value 0.89).

Scale-up preview — the hard domain and official iGSM

Claim: Accuracy degrades sharply on 10–18-step problems; counterfactual preference learning, capacity, and search depth provide complementary gains.

hard domain (14–24 vars, 10–18 steps)	@opt	@slack2	look-2	look-4
reference recipe (`combo`, 9M)	0.000	0.070	0.035	0.175
geometry-distilled + cross-horizon calibration (9M)	**0.120**	**0.445**	0.105	0.150
same recipe, 35M	0.090	0.410	0.195	**0.315**
token LM (8M, outcome-candidate score)	0.010	—	—	—

Capacity only pays *through search* (35M: 0.09 greedy → 0.32 at look-4). And on **100%-faithful iGSM** (official facebookresearch generator vendored; our env drives the reference renderer; solutions pass the official checker 60/60 med, 99/100 hard): the reference recipe is near-random (0.245 vs 0.20; probes healthy — the energy fails, the pre-ranking signature), 60 epochs don't help (0.235), and **counterfactual ranking is again the fix: 0.410 @optimal / 0.735 @slack-2**, and capacity stacks on it (35M: 0.460/0.765, value probe 0.53→0.93) — the recipe's central mechanism transfers to the official domain, with the remaining gap to oracle as the next phase's frontier.

Removing symbolic preference labels: geometric ranking

Claim: Geometric-advantage ranking (GAR) replaces remaining-step and binary preference targets with an ordering induced entirely by encoded outcomes. At one visited state, execute the trace action and K alternative feasible actions in cloned environments. Encode each true next sentence with the EMA encoder and measure its distance to the encoded terminal trace state:

$$g_i = \|\mathrm{LN}(\bar{s}_{t+1,i}^{\mathrm{true}}) - \mathrm{LN}(\bar{s}_T)\|_1, \qquad E_i = V(F(s_t, a_i), s_0).$$

GAR trains $E_i < E_j$ whenever $g_i < g_j$, using only pairwise margins. It uses no remaining-step count and no symbolic relevance flag.

Boundary of the claim. GAR still enumerates feasible actions, renders counterfactual next sentences, and uses intent phrases. The current GAR jobs also retain operation-type displacement decoding. It is therefore *preference-label-free*, not fully action-annotation-free. Two GAR variants are running; results are not yet included in the tables.

Full variational JEPA over an undifferentiated sentence stream

Claim: The new model removes intent phrases entirely and represents uncertainty in both the unobserved transition and the next latent state. Prompt and solution sentences form one causal stream; every adjacent pair is a training transition. Following VJEPA (arXiv:2601.14354), the EMA target encoder defines a diagonal-Gaussian inference distribution, a target sample is drawn, and the predictor assigns it likelihood:

$$
\begin{aligned}
q_\phi(z_{t+1} \mid x_{\le t+1}) &= \mathcal{N}(\bar{\mu}_{t+1}, \operatorname{diag}\sigma^2_{q,t+1}),\\
q_\psi(u_t \mid s_t, z_{t+1}) &= \mathcal{N}(\mu^u_q, \operatorname{diag}(\sigma^u_q)^2), \quad p_\omega(u_t \mid s_t) = \mathcal{N}(\mu^u_p, \operatorname{diag}(\sigma^u_p)^2),\\
p_\theta(z_{t+1} \mid s_t, u_t) &= \mathcal{N}(\mu^z_\theta, \operatorname{diag}(\sigma^z_\theta)^2),\\
\mathcal{L} &= -\log p_\theta(z^{(q)}_{t+1} \mid s_t, u^{(q)}_t) + \beta_z\, D_{\mathrm{KL}}(q_\phi \| \mathcal{N}(0, I)) + \beta_u\, D_{\mathrm{KL}}(q_\psi \| p_\omega).
\end{aligned}
$$

Both predictor and target variances are learned. There is no token decoder, intent/outcome split, symbolic action target, or feasible-action interface. This first experiment evaluates representation dynamics and calibration, not task planning. Stylized and official-iGSM runs are active.

Latent displacement decoding with unobserved actions: factorial test

Claim: EMA targets require an explicit distribution regularizer here: VICReg or SIGReg preserves high-rank states, whereas removing EMA produces collapsed or strongly low-rank states in every cell. Delta-JEPA observes a_t and minimizes $\|D(s_{t+1} - s_t) - a_t\|_2^2$. Our action-free extension instead uses $\|D(s_{t+1} - s_t) - \mathrm{sg}\,\mu_q^u\|_2^2$. The latter is a consistency constraint: q_ψ and D can agree on an uninformative code, so we report posterior-mean standard deviation and effective rank rather than transferring Delta-JEPA's anti-collapse claim.

target encoder	no state regularizer	VICReg	SIGReg (LeJEPA)
EMA target	2.9 / 1.9	**125.1 / 9.2**	**128.8 / 12.8**
shared online target (no EMA/stop-grad)	collapse†	2.4 / 2.2	16.8 / 7.7

Each entry is state effective rank / posterior-action-mean effective rank (maxima 256 / 16) after the same 30k-problem, 10-epoch pilot. †State std. $= 9\times10^{-4}$ and action-mean std. $= 2.5\times10^{-3}$; effective rank is misleading at this scale. SIGReg follows LeJEPA's random-projection normality test. These are collapse diagnostics, not evidence that the inferred action code is semantically grounded.

Experiments active on July 13, 2026

question	runs	status
Can geometric outcomes replace symbolic preference labels?	GAR + GAR-pure	training
How many official-iGSM counterfactuals are needed?	$K = 1$ and $K = 4$ (with $K = 2$ reference)	training
Does counterfactual transition data help without ranking?	outcome prediction, $K = 1, 2, 4$	training
Can fully latent actions model a sentence stream probabilistically?	stylized + official iGSM	training
How do EMA, VICReg, and SIGReg interact with latent LDAD?	2×3 factorial	completed; diagnostics above

Only the completed LDAD pilot diagnostics are reported above. The other rows remain active and should enter result tables only after checkpoint, validation, and the relevant planning or counterfactual-audit artifact exist.

Honest limitations & what this is not

- Most ablations use the stylized flat-DAG generator. Transfer is tested on the vendored official iGSM generator, but not yet at broad language variation or natural-text scale.
- The reported planning records use observed intent phrases and an oracle feasible-action enumerator. The new sentence-stream VJEPA removes both interfaces, but does not yet support task planning.
- Geometry-distilled values remove scalar remaining-step labels but the best completed variant still uses binary relevance and symbolic counterfactual ordering. GAR removes those preference labels but still uses environment-generated counterfactual outcomes.
- Value labels shape only the value head unless `valgrad`; operation CE in the hybrid displacement loss is explicit symbolic action-type supervision.
- Historical data-scale artifacts reused fixed per-index corpora and changed optimizer steps with corpus size. Fresh-per-epoch generation is fixed for new experiments; the old diversity claim needs a matched rerun.
- Text "generation" is action selection + template rendering; a learned renderer is orthogonal to the world model.

Next steps after the active runs

- **Resolve the causal question**: compare $K = 1, 2, 4$ counterfactual outcome prediction against matched preference-learning runs, with identical updates and fresh problems.
- **Audit latent actions**: posterior-mean variance/effective rank, transition clustering, uncertainty calibration, and held-out next-sentence likelihood for every EMA/regularizer cell.
- **Recover planning without action names**: learn a proposal over latent u_t and a separate observation-to-action execution mechanism; do not nearest-match against an oracle intent list in the final system.
- **Scale only after the faithful verdict**: use the official iGSM K sweep to select counterfactual density, then increase capacity and search horizon with a compute-matched protocol.
- **Broaden text**: paraphrases, larger moduli, noisy statements, and ultimately non-templated corpora; retain counterfactual audits.